Decoding Earth's Aerobiomes: Implications for Astrobiology and the Search for Extraterrestrial Life

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Introduction

Modern sequencing techniques have revolutionized our perception of microorganisms, revealing their existence even in the harsh conditions of Earth's atmosphere and hinting at life's potential elsewhere in the universe. By studying these atmospheric microorganisms, we not only enhance our understanding of Earth's ecosystem, but also devise strategies to explore life on other planets, an endeavor supported by intriguing discoveries such as phosphine on Venus, which suggests possible microbial existence.

Methodology

FASTQ files (NCBI & NASA databases)

↓ Quality control

FastQC

MultiQC

↓ Remove adapters

Trimmomatic

↓ Quality control

↓ Taxonomic classification

Kraken2

↓ Visualisation of results

Jupytool interactive tool

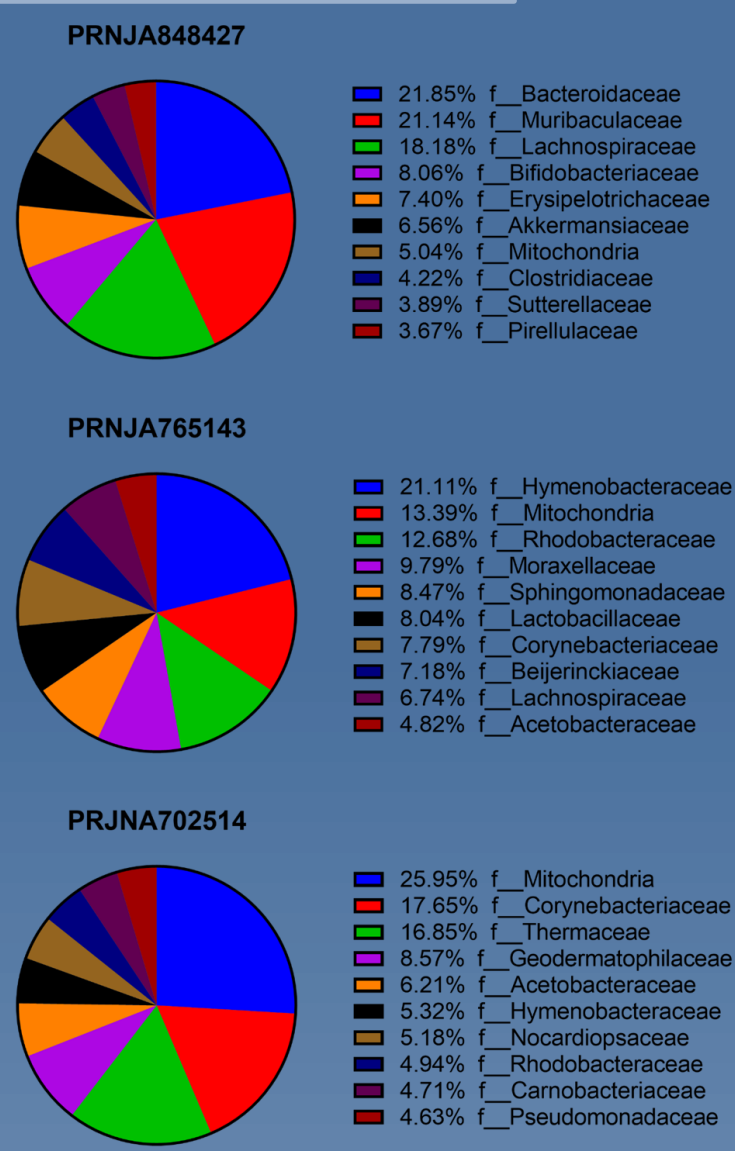
Our project is based on data in public repositories. In our preliminary study, we used the projects:

- **PRJNA702514**[1] (Arizona State University, 114 SRA)
- **PRJNA765143**[2] (East Mediterranean, 145 SRA)
- **PRJNA848427**[3] (Queensland and New South Wales, 149 SRA)
- **NASA_OSD-170**[4] (United States)

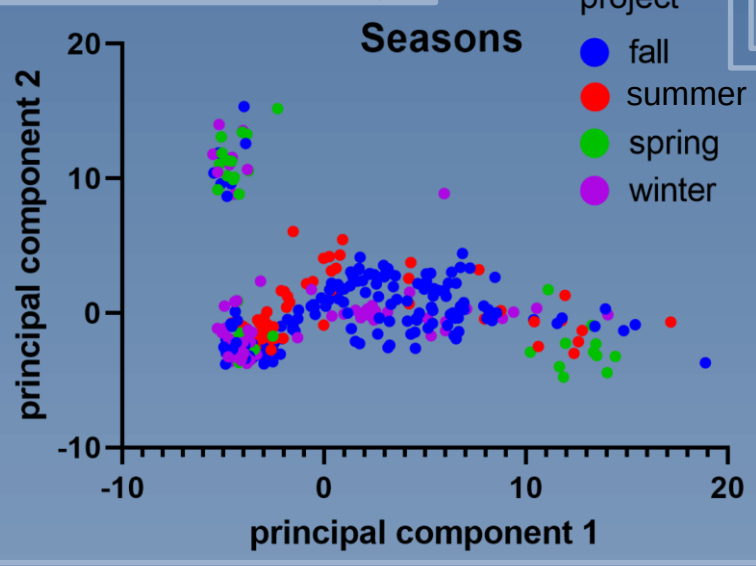
PRJNA samples collected several meters above the ground, NASA atmosphere samples up to 11 km.

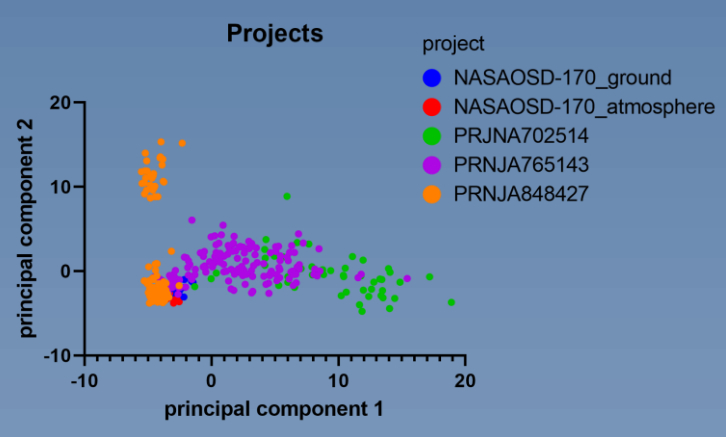
Galaxy NCBI

Microbial composition

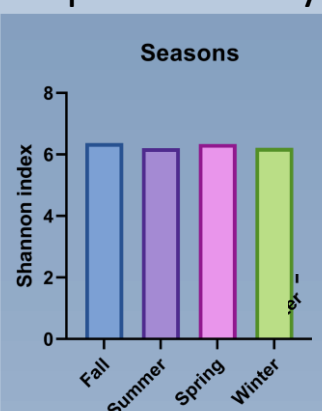


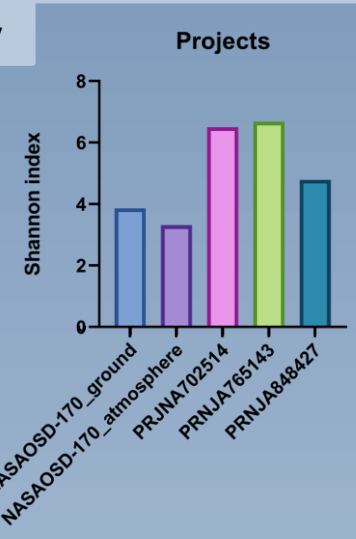
Seasonal variability





Alpha-diversity





Results

Earth's atmosphere is composed primarily of nitrogen, oxygen and traces of other gases, and plays a crucial role in maintaining suitable conditions for life. Conditions vary with altitude, including decreasing pressure and temperature as one ascends. These changes are accompanied by phenomena like weather patterns, the occurrence of air pollution and the presence of aerosols and particulate matter. Our preliminary results suggest that the greatest difference in diversity occurs between different ecosystems. There are also differences between seasons. Bacteria occur in various ratios between different time intervals. The identified taxa include not only environmental bacteria, but also families comprising bacteria potentially pathogenic to humans.

11 km altitude

At an altitude of 11 km, within the tropopause, temperatures average around -50 to -60 °C, atmospheric pressure drops to approximately a quarter of that at sea level, and oxygen availability decreases due to lower pressure, posing harsh conditions for life. However, advanced scientific methods have identified resilient microorganisms at these altitudes, offering insight into how life adapts to extreme conditions and hinting at the potential for extraterrestrial existence.

10 m altitude

At 10 meters above the ground, the atmosphere still closely resembles the conditions at surface level, with temperatures reflecting local weather patterns, high oxygen levels, and standard atmospheric pressure, creating a suitable environment for a diverse array of microorganisms. These microorganisms, part of the aerobiome, can be found both free-floating and attached to aerosol particles, playing vital roles in a range of processes from disease transmission to environmental nutrient cycling.

Ground

The atmosphere near the ground offers an environment rich in oxygen and maintains a consistent pressure and temperature, suitable for the existence and thriving of a multitude of microorganisms. These microscopic beings, being key components of the biosphere, significantly contribute to various vital processes like decomposition, disease transmission, and nutrient cycling.

Aerobiome

To study bioaerosol microorganisms on celestial bodies where DNA sequencing might be unavailable, we propose using metabolomic profiling via spectrophotometry, offering an indirect, less resource-intensive identification method. By analyzing metabolites, products of microbial processes, we gain insights into the functioning of these microbial ecosystems. This innovative approach paves the way for significant astrobiological exploration, extending our understanding of life within extreme environments.

References

[1] https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA702514

[2] Gat, D. *et al.* (2021) 'Size-Resolved Community Structure of Bacteria and Fungi Transported by Dust in the Middle East', *Frontiers in Microbiology*, 12. Available at: https://doi.org/10.3389/fmicb.2021.744117

[3] https://www.ncbi.nlm.nih.gov/bioproject/?term=PRJNA848427

[4] https://osdr.nasa.gov/bio/repo/data/studies/OSD-170